

Our speedup method can only compute the best match because of nearest-neighbor searching

Only train split

The base method (PS*_vanilla) is slow, so only the train split is considered for now.

Results. GunPoint (Train split)

It requires a pairwise distance matrix.

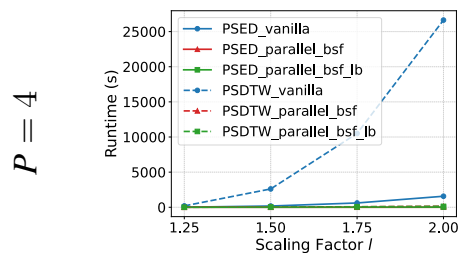
Guided distance

Bold refers to worsen w.r.t the corresponding distance measure

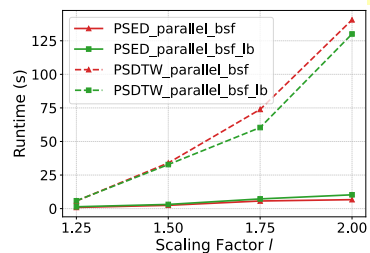
P	l	ED		DTW [19]		ADTW [37]		DDTW [38]		shapeDTW [39]		WDDTW [40]		WDTW [40]		PSED		PSDTW	
		$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$
2	1.25	0.30	0.56	0.82	1.00	0.54	0.84	0.84	0.92	0.96	1.00	0.88	0.96	0.82	0.98	0.98	1.00	0.86	1.00
	1.50	0.14	0.36	0.88	1.00	0.38	0.64	0.78	0.94	1.00	1.00	0.80	0.92	0.88	0.96	1.00	1.00	<u>0.96</u>	1.00
	1.75	0.16	0.34	0.82	1.00	0.38	0.66	0.64	0.80	0.90	1.00	0.66	0.80	0.82	0.96	1.00	1.00	<u>0.94</u>	1.00
	2.00	0.12	0.24	0.84	0.98	0.34	0.66	0.72	0.88	<u>0.96</u>	1.00	0.68	0.86	0.82	0.98	1.00	1.00	<u>0.96</u>	1.00
3	1.25	0.30	0.50	0.84	0.92	0.70	0.88	0.80	0.92	0.96	0.98	0.84	0.94	0.86	0.98	<u>0.94</u>	1.00	0.88	0.96
	1.50	0.10	0.28	0.84	0.96	0.60	0.78	0.66	0.86	<u>0.90</u>	1.00	0.72	0.86	0.84	0.98	0.96	1.00	0.86	0.96
	1.75	0.10	0.28	<u>0.88</u>	1.00	0.42	0.78	0.72	0.88	0.76	0.94	0.74	0.88	0.88	1.00	1.00	1.00	<u>0.88</u>	0.98
	2.00	0.02	0.12	<u>0.86</u>	0.94	0.38	0.52	0.60	0.78	0.70	0.94	0.60	0.80	0.82	0.92	0.98	1.00	<u>0.86</u>	0.98
4	1.25	0.34	0.50	0.70	0.86	0.68	0.88	0.60	0.78	0.70	0.90	0.64	0.80	0.70	0.86	0.86	0.94	<u>0.72</u>	0.88
	1.50	0.22	0.38	0.64	0.80	0.60	0.92	0.52	0.64	0.68	0.90	0.52	0.66	0.64	0.82	0.86	0.98	<u>0.72</u>	0.82
	1.75	0.16	0.30	0.72	0.92	0.56	0.80	0.60	0.78	0.56	0.80	0.56	0.80	0.70	0.88	0.92	1.00	<u>0.80</u>	0.94
	2.00	0.06	0.12	0.58	0.78	0.50	0.72	0.46	0.64	0.50	0.82	0.50	0.64	0.56	0.82	0.88	0.98	<u>0.64</u>	0.88

P	l	DTW ^{PSED}		ADTW ^{PSED}		DDTW ^{PSED}		shapeDTW ^{PSED}		WDDTW ^{PSED}		WDTW ^{PSED}	
		$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$	$P@1$	$P@3$
2	1.25	0.86	1.00	0.98	1.00	0.72	0.92	0.98	1.00	0.74	0.96	0.86	1.00
	1.50	0.88	1.00	1.00	1.00	0.60	0.90	0.98	1.00	0.64	0.90	0.92	1.00
	1.75	0.94	1.00	1.00	1.00	0.84	0.94	1.00	1.00	0.86	0.96	0.94	1.00
	2.00	0.98	1.00	1.00	1.00	0.60	0.78	0.98	1.00	0.66	0.80	0.98	1.00
3	1.25	0.82	0.96	0.94	1.00	0.74	0.90	0.92	0.98	0.76	0.90	0.82	0.96
	1.50	0.86	1.00	0.96	1.00	0.72	0.90	0.94	1.00	0.80	0.90	0.88	1.00
	1.75	0.88	1.00	1.00	1.00	0.68	0.90	1.00	1.00	0.70	0.94	0.90	1.00
	2.00	0.86	0.96	0.98	1.00	0.54	0.86	0.96	1.00	0.62	0.86	0.88	0.96
4	1.25	0.74	0.88	0.86	0.94	0.78	0.80	0.80	0.92	0.78	0.82	0.74	0.88
	1.50	0.72	0.84	0.86	0.98	0.68	0.84	0.84	0.96	0.66	0.84	0.72	0.86
	1.75	0.74	0.96	0.92	1.00	0.54	0.90	0.90	1.00	0.56	0.88	0.74	0.96
	2.00	0.66	0.90	0.88	0.98	0.58	0.82	0.88	0.98	0.62	0.80	0.66	0.90

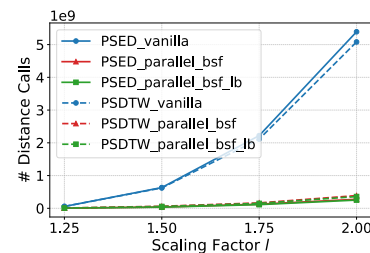
Running time



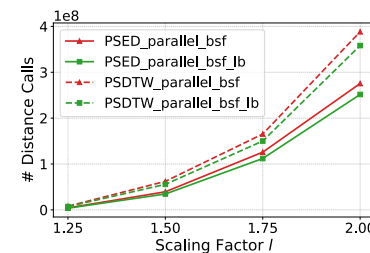
(i)



(j)



(k)



(l)

The lower bounding can speed up PSDTW but not PSED

Because the overhead of calculating the lower bound outweighs the time saved by pruning

Results: 10 datasets (Train split)

Dataset	ED		DTW		ADTW		DDTW		shapeDTW		WDDTW		WDTW		PSED	
	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$
SonyAIBORobotSurface1	0.30	0.70	<u>0.90</u>	1.00	1.00	1.00	0.80	0.85	0.50	0.70	0.80	0.85	1.00	1.00	1.00	1.00
ECG200	0.12	0.20	0.70	0.81	<u>0.78</u>	0.83	0.60	0.74	0.59	0.76	0.59	0.73	0.70	0.80	0.81	0.88
MedicalImages	0.08	0.19	0.63	0.78	<u>0.69</u>	0.85	0.52	0.68	0.57	0.75	0.51	0.68	0.62	0.78	0.76	0.88
CBF	0.53	0.67	0.83	1.00	<u>0.90</u>	1.00	0.73	0.87	0.73	0.90	0.77	0.90	<u>0.90</u>	1.00	1.00	1.00
SwedishLeaf	0.07	0.12	0.82	0.92	<u>0.84</u>	0.93	0.70	0.83	0.78	0.91	0.69	0.83	0.83	0.92	0.97	0.99
Plane	0.09	0.14	0.50	0.74	<u>0.59</u>	0.78	0.38	0.64	0.53	0.79	0.37	0.61	0.49	0.75	0.75	0.93
PowerCons	0.26	0.43	<u>0.77</u>	0.98	0.80	1.00	<u>0.77</u>	0.98	<u>0.77</u>	0.99	<u>0.77</u>	0.99	<u>0.77</u>	0.99	0.80	1.00
GunPoint	0.10	0.28	<u>0.84</u>	0.96	0.60	0.78	<u>0.66</u>	0.86	<u>0.90</u>	1.00	<u>0.72</u>	0.86	<u>0.84</u>	0.98	0.96	1.00
Adiac	0.01	0.02	<u>0.37</u>	0.50	0.12	0.21	0.32	0.42	<u>0.27</u>	0.41	0.31	0.41	0.36	0.50	0.57	0.71
Epilepsy	0.18	0.26	<u>0.74</u>	0.88	<u>0.80</u>	0.91	0.65	0.82	0.38	0.47	0.58	0.79	0.70	0.83	0.83	0.91

Dataset	DTW ^{PSED}		ADTW ^{PSED}		DDTW ^{PSED}		shapeDTW ^{PSED}		WDDTW ^{PSED}		WDTW ^{PSED}	
	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$	$\overline{P@1}$	$\overline{P@3}$
SonyAIBORobotSurface1	0.95	1.00	1.00	1.00	0.65	0.80	1.00	1.00	0.75	0.80	0.95	1.00
ECG200	0.73	0.84	0.80	0.88	0.54	0.68	0.81	0.88	0.53	0.69	0.73	0.84
MedicalImages	0.66	0.78	0.76	0.88	0.51	0.69	0.74	0.88	0.51	0.70	0.66	0.79
CBF	0.90	1.00	0.93	1.00	0.73	0.93	0.93	0.97	0.70	0.93	0.90	1.00
SwedishLeaf	0.82	0.92	0.97	0.99	0.72	0.87	0.97	0.99	0.73	0.88	0.82	0.92
Plane	0.60	0.80	0.75	0.94	0.50	0.67	0.74	0.90	0.50	0.69	0.59	0.80
PowerCons	0.78	0.99	0.79	1.00	0.79	1.00	0.79	1.00	0.79	1.00	0.78	1.00
GunPoint	0.86	1.00	0.96	1.00	0.72	0.90	0.94	1.00	0.80	0.90	0.88	1.00
Adiac	0.33	0.45	0.57	0.71	0.21	0.34	0.53	0.70	0.20	0.33	0.34	0.45
Epilepsy	0.77	0.88	0.78	0.88	0.66	0.79	0.74	0.89	0.66	0.82	0.80	0.89

Final results

10 datasets (Train split) Running time

Name	PSED_vanilla	PSED_parallel_bsf	% distance calls pruned	Speedup
	Time (s)	Time (s)		
SonyAIBORobotSurface1	0.745	0.089	77.69%	8.40X
ECG200	96	3	90.74%	33.62X
MedicalImages	2108	36	96.44%	59.26X
CBF	41	1	74.75%	32.52X
SwedishLeaf	9785	98	96.19%	99.81X
Plane	648	5	96.66%	126.51X
PowerCons	2058	40	83.64%	51.81X
GunPoint	211	2	92.25%	99.33X
Adiac	19547	109	96.66%	178.72X
Epilepsy	7510	268	43.46%	28.01X

10 datasets (Train split + Test split)

Dataset	ED	DTW	ADTW	DDTW	shapeDTW	WDDTW	WDTW	PSED	Time (s)
SonyAIBORobotSurface1	0.05	0.51	<u>0.61</u>	0.45	0.20	0.44	0.51	0.67	65
ECG200	0.07	0.65	<u>0.77</u>	0.52	0.56	0.50	0.66	0.79	10
MedicalImages	0.06	0.52	<u>0.62</u>	0.39	0.50	0.38	0.51	0.68	281
CBF	0.09	0.55	<u>0.63</u>	0.40	0.31	0.40	0.55	0.70	640
SwedishLeaf	0.05	0.73	<u>0.76</u>	0.58	0.68	0.59	0.74	0.95	447
Plane	0.07	0.38	<u>0.40</u>	0.24	<u>0.46</u>	0.24	0.37	0.68	19
PowerCons	0.15	0.61	<u>0.63</u>	0.59	0.61	0.59	0.61	0.64	129
GunPoint	0.06	0.59	<u>0.23</u>	0.53	<u>0.67</u>	0.54	0.61	0.88	22
Adiac	0.01	<u>0.34</u>	0.09	0.28	<u>0.25</u>	0.27	<u>0.34</u>	0.53	417
Epilepsy	0.16	0.71	<u>0.76</u>	0.60	0.37	0.54	0.66	0.82	1113

Length l

Length of immediate subsequence

Dataset	Data				Forecasting Task		
	N	n	rate	start date	h	$ T_{\text{train}} $	$ T_{\text{test}} $
Electricity [22]	70	26,136	hourly	2012-01-01, 00:00	24	25,968	168
Traffic [22]	90	10,560	hourly	2012-01-01, 00:00	24	10,392	168
PeMSD7 [22]	228	12,672	/5 mins	2012-05-01, 00:00	9	11,232	1,440
Exchange-Rate [1]	8	7,536	daily	1990-01-01, 00:00	24	6,048	1,488

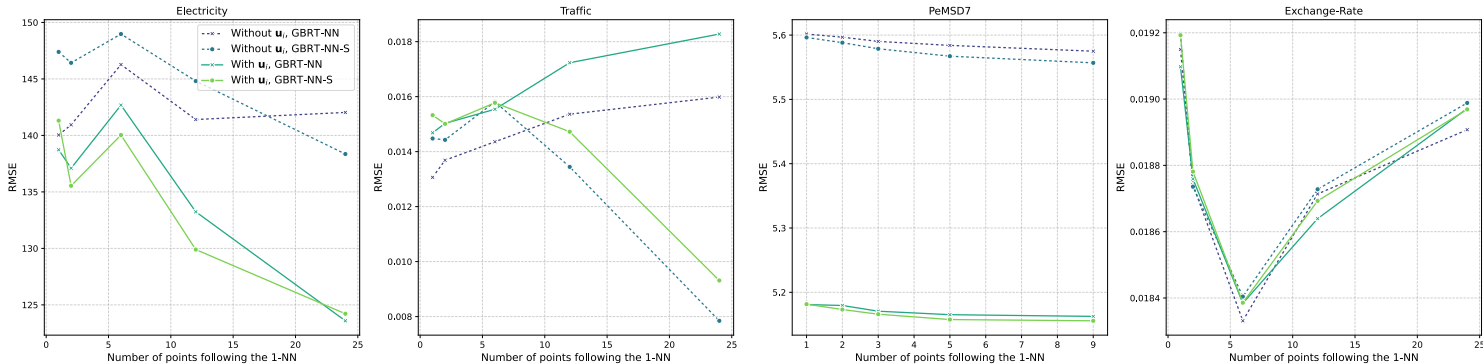


FIGURE 3. Effect of different lengths l of the immediate subsequence (i.e., Number of points following the 1-NN) on RMSE.

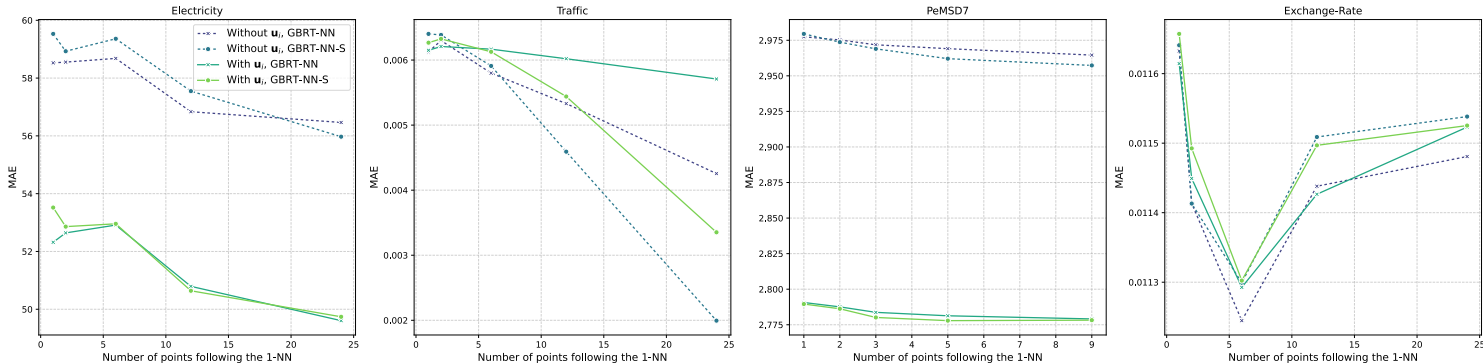


FIGURE 4. Effect of different lengths l of the immediate subsequence (i.e., Number of points following the 1-NN) on MAE.

Basic Results

Without known input time-covariates

Baseline

This baseline has been shown to be comparable with other complex methods, such as deep learning

Dataset	Data				Forecasting Task		
	N	n	rate	start date	h	$ T_{\text{train}} $	$ T_{\text{test}} $
Electricity [22]	70	26,136	hourly	2012-01-01, 00:00	24	25,968	168
Traffic [22]	90	10,560	hourly	2012-01-01, 00:00	24	10,392	168
PeMSD7 [22]	228	12,672	/5 mins	2012-05-01, 00:00	9	11,232	1,440
Exchange-Rate [1]	8	7,536	daily	1990-01-01, 00:00	24	6,048	1,488

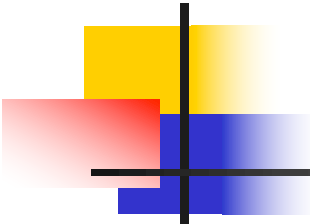
Using $l = h$

Using $l = 6$

Dataset	Metric	GBRT	GBRT-NN	GBRT-NN-S	GBRT-NN ⁺	GBRT-NN ⁺ -S	GBRT-NN ⁺⁺	GBRT-NN ⁺⁺ -S
Electricity	RMSE	136.254	142.035	<u>138.354</u>	146.399	141.019	143.645	142.546
	WAPE	0.103	<u>0.101</u>	0.100	0.102	0.100	<u>0.101</u>	0.100
	MAE	57.929	56.464	55.972	56.898	56.019	56.401	<u>55.990</u>
Traffic	RMSE	<u>0.012</u>	0.016	0.008	0.016	0.008	0.015	0.008
	WAPE	0.108	0.079	<u>0.037</u>	0.075	<u>0.037</u>	0.064	0.035
	MAE	0.006	0.004	0.002	0.004	0.002	<u>0.003</u>	0.002
PeMSD7	RMSE	5.610	5.575	5.557	5.571	<u>5.554</u>	5.564	5.547
	WAPE	0.051	0.051	0.051	0.051	0.051	0.051	0.051
	MAE	2.984	2.965	2.957	2.963	<u>2.955</u>	2.959	2.951
Exchange-Rate	RMSE	<u>0.020</u>	0.019	0.019	0.019	0.019	0.021	0.021
	WAPE	<u>0.016</u>	0.015	0.015	0.015	0.015	<u>0.016</u>	<u>0.016</u>
	MAE	<u>0.012</u>	0.011	<u>0.012</u>	<u>0.012</u>	<u>0.012</u>	<u>0.012</u>	<u>0.012</u>

With known input time-covariates

Dataset	Metric	GBRT	GBRT-NN	GBRT-NN-S	GBRT-NN ⁺	GBRT-NN ⁺ -S	GBRT-NN ⁺⁺	GBRT-NN ⁺⁺ -S
Electricity	RMSE	132.669	123.591	<u>124.215</u>	130.869	130.428	125.822	129.078
	WAPE	0.093	0.089	0.089	0.090	0.090	0.089	0.090
	MAE	52.023	49.606	<u>49.739</u>	50.385	50.671	50.112	50.342
Traffic	RMSE	0.014	0.018	0.009	0.019	0.009	0.019	0.009
	WAPE	0.109	0.106	0.062	0.102	<u>0.061</u>	0.096	0.058
	MAE	0.006	0.006	0.003	0.006	0.003	<u>0.005</u>	0.003
PeMSD7	RMSE	5.191	5.163	5.156	5.161	<u>5.148</u>	5.152	5.145
	WAPE	0.048	0.048	0.048	0.048	0.048	0.048	0.048
	MAE	2.796	2.779	2.778	2.780	<u>2.773</u>	2.774	2.771
Exchange-Rate	RMSE	<u>0.020</u>	0.019	0.019	0.019	0.019	0.021	0.021
	WAPE	<u>0.016</u>	0.015	0.015	0.015	0.015	<u>0.016</u>	<u>0.016</u>
	MAE	0.012	0.012	0.012	0.012	0.012	0.012	0.012



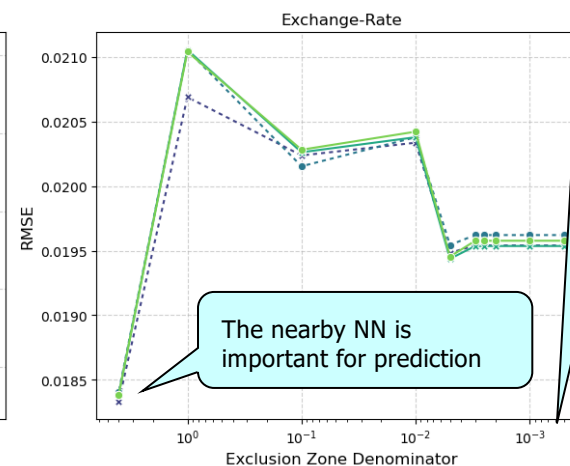
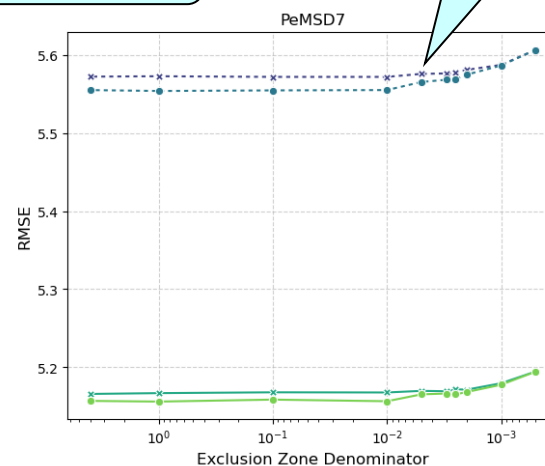
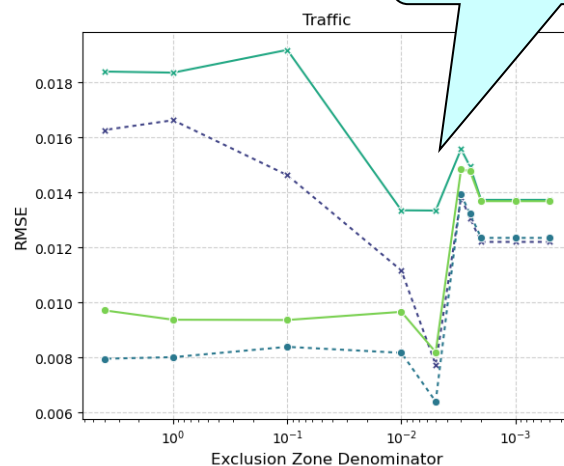
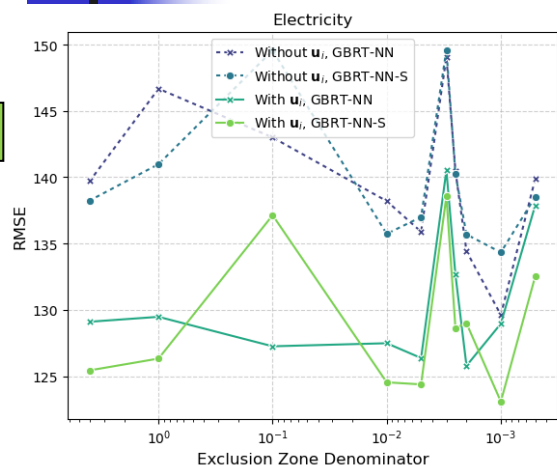
Final Results

k-nearest neighbor

Dataset	Without time-covariates		With time-covariates	
Electricity	GBRT (136.254, 0.103, 57.929),	GBRT-3NN⁺⁺-S (145.004, 0.099, 54.895)	GBRT-NN (123.591, 0.089, 49.606),	GBRT-2NN⁺⁺-S (129.694, 0.088, 48.822)
Traffic		GBRT-NN⁺⁺-S (0.008, 0.035, 0.002)		GBRT-NN⁺⁺-S (0.009, 0.058, 0.003)
PeMSD7		GBRT-3-NN⁺⁺-S (5.514, 0.050, 2.932)		GBRT-3NNC⁺⁺-S (5.090, 0.047, 2.743)
Exchange-Rate		GBRT-NN (0.019, 0.015, 0.011)		GBRT-NN (0.019, 0.015, 0.012)

Study on the length (n) of a time series used to find the NN (2/2)

RMSE

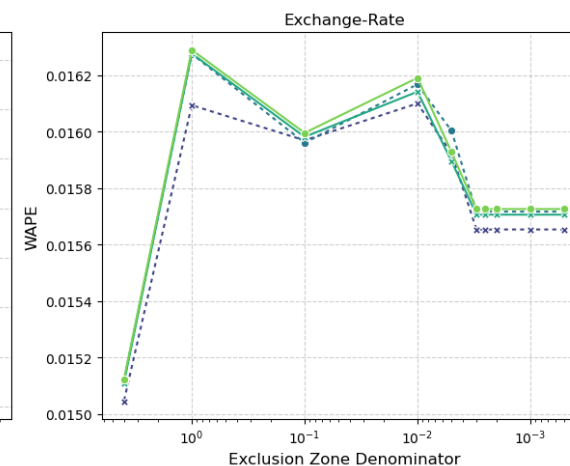
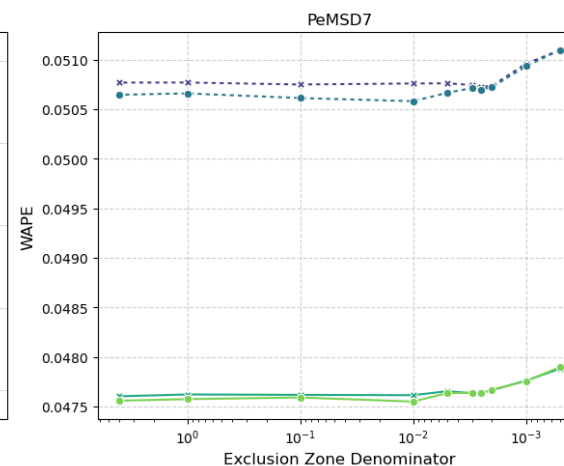
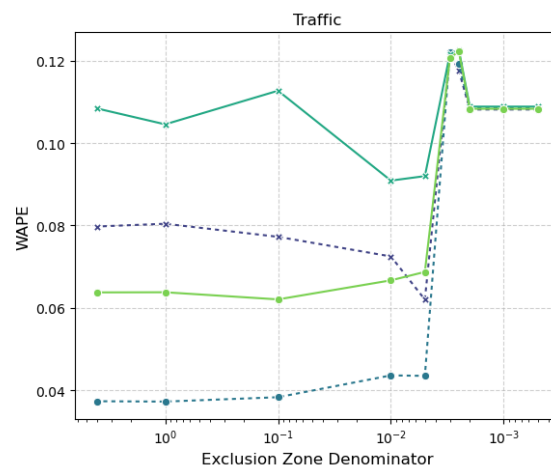
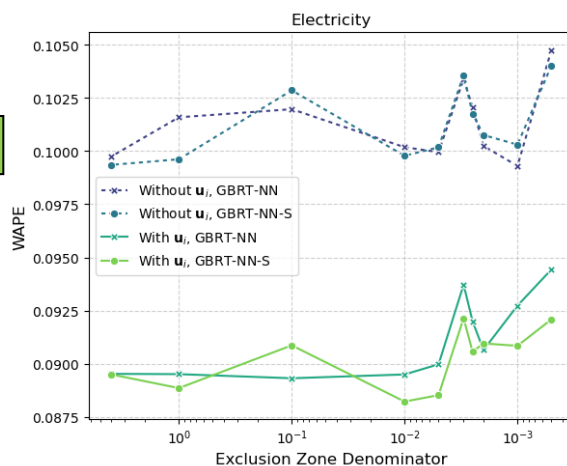


It means the further-away NN is more useful for prediction

The NN everywhere is similar for prediction

Denominator **smaller** -> Exclusion zone **larger** -> n used for finding left-NN **shorter**

MAE



The nearby NN is important for prediction

Study on effect of m

Query length

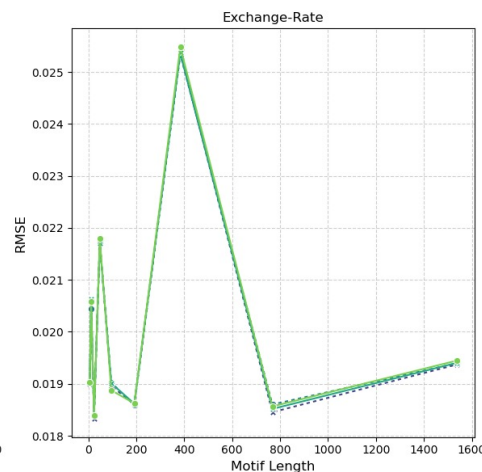
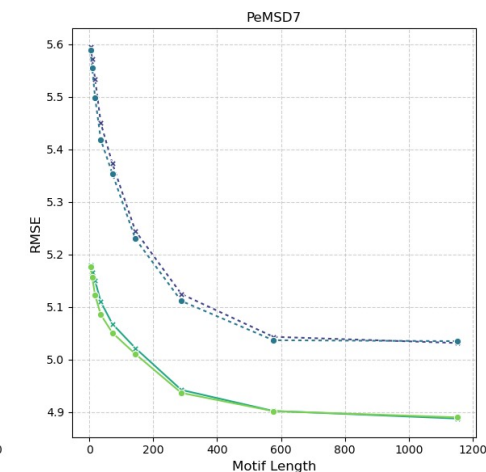
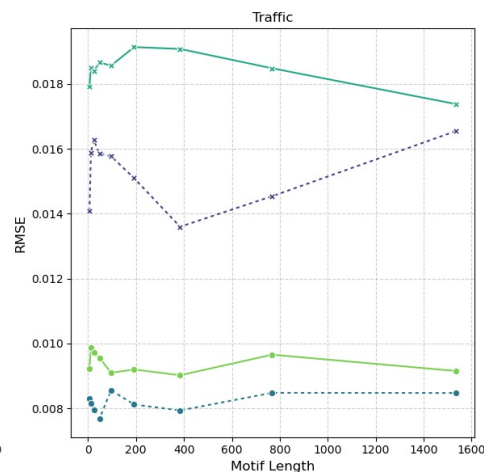
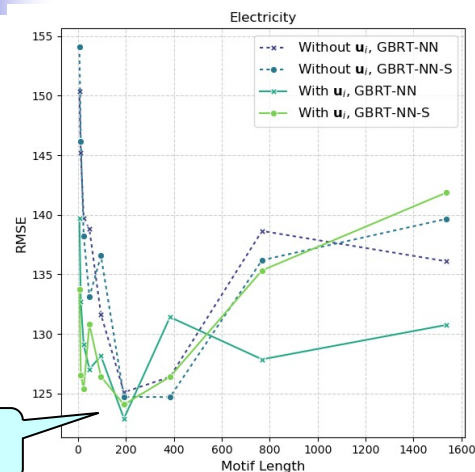
Dataset	Data				Forecasting Task		
	N	n	rate	start date	h	$ T_{\text{train}} $	$ T_{\text{test}} $
Electricity [22]	70	26,136	hourly	2012-01-01, 00:00	24	25,968	168
Traffic [22]	90	10,560	hourly	2012-01-01, 00:00	24	10,392	168
PeMSD7 [22]	228	12,672	/5 mins	2012-05-01, 00:00	9	11,232	1,440
Exchange-Rate [1]	8	7,536	daily	1990-01-01, 00:00	24	6,048	1,488

We tested for 4, 9, 18, 36, 72, 144, 288, 576, 1152

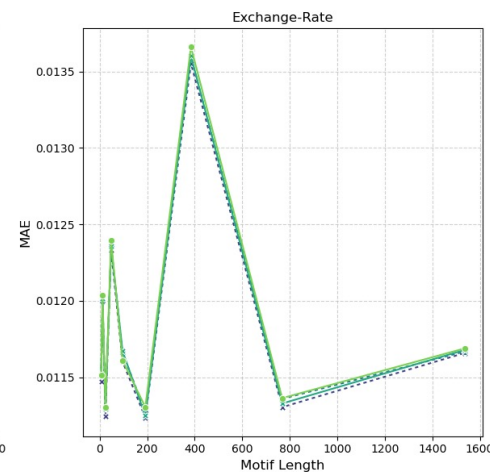
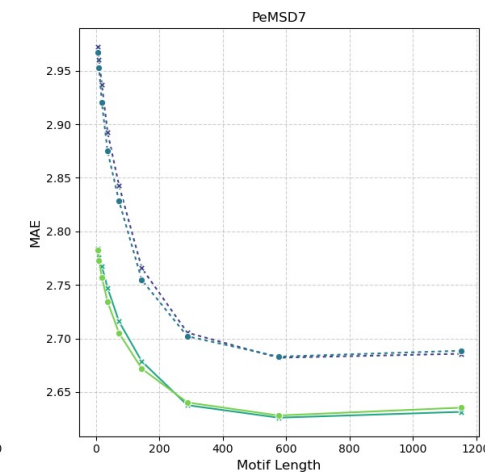
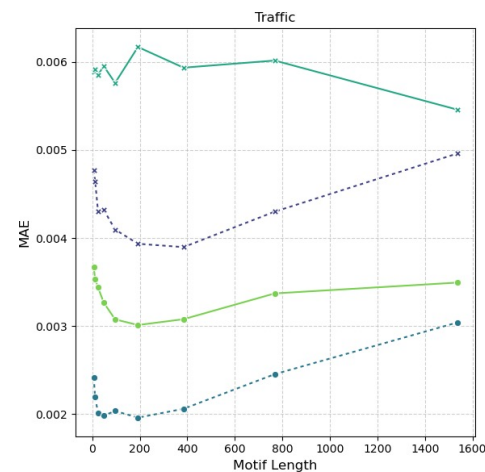
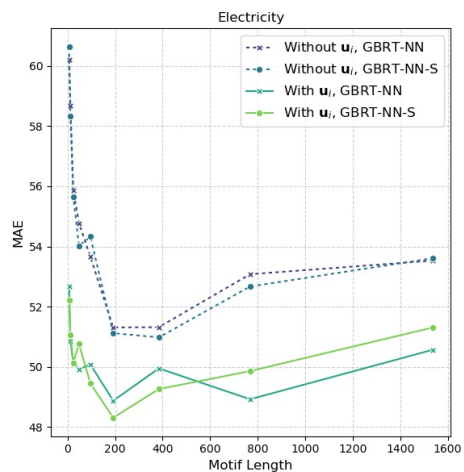
We tested for 6, 12, 24, 48, 96, 192, 384, 768, 1536

RMSE

It is best at here



MAE



We first verify what a good combination of the channel is

We use single-value mapping.

Performance across the channel

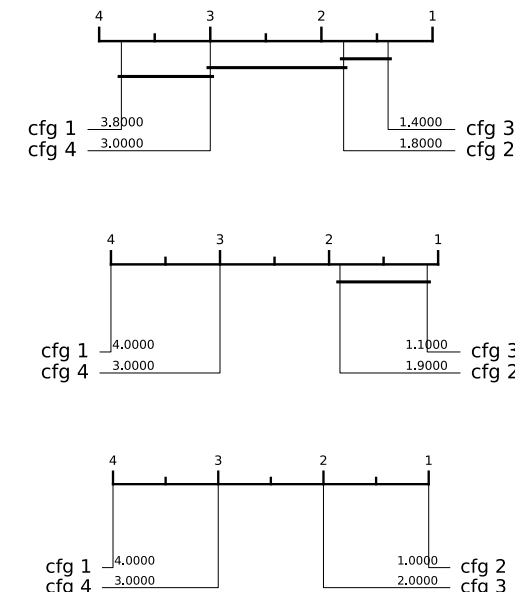
Only sequence data

Secondary structure is a good covariate

Standalone is better.

	Classifier	5p arm					3p arm					multi-class				
		Acc	Sp	Sn	F1	MCC	Acc	Sp	Sn	F1	MCC	Acc	Sp	Sn	F1	MCC
Baseline (cfg 1)	ROCKET	0.781	0.743	0.819	0.789	0.563	0.790	0.773	0.807	0.793	0.580	0.717	0.838	0.685	0.700	0.538
	MiniROCKET	0.755	0.728	0.782	0.762	0.512	0.788	0.781	0.794	0.789	0.576	0.685	0.823	0.653	0.662	0.486
	MultiROCKET	0.784	0.767	0.801	0.787	0.569	0.803	0.792	0.814	0.805	0.606	0.691	0.830	0.667	0.672	0.501
	Hydra	0.830	0.800	0.860	0.835	0.663	0.808	0.797	0.820	0.810	0.617	0.731	0.844	0.696	0.712	0.560
	MultiROCKET-Hydra	0.796	0.778	0.815	0.800	0.594	0.807	0.767	0.816	0.808	0.614	0.701	0.836	0.681	0.686	0.520
Baseline + Secondary Structure (cfg 2)	ROCKET	0.847	0.832	0.862	0.849	0.695	0.855	0.842	0.868	0.857	0.711	0.836	0.907	0.828	0.833	0.736
	MiniROCKET	0.825	0.807	0.843	0.827	0.652	0.822	0.802	0.843	0.826	0.646	0.823	0.900	0.812	0.818	0.715
	MultiROCKET	0.812	0.803	0.822	0.814	0.626	0.824	0.809	0.839	0.826	0.649	0.796	0.888	0.791	0.792	0.673
	Hydra	0.845	0.816	0.873	0.849	0.691	0.846	0.817	0.874	0.850	0.693	0.830	0.901	0.814	0.826	0.724
	MultiROCKET-Hydra	0.817	0.809	0.826	0.819	0.635	0.825	0.816	0.834	0.826	0.652	0.803	0.891	0.798	0.800	0.684
Baseline + Base-pair probability (Standalone) (cfg 3)	ROCKET	0.842	0.828	0.855	0.844	0.684	0.855	0.856	0.854	0.855	0.710	0.795	0.885	0.783	0.789	0.670
	MiniROCKET	0.817	0.820	0.814	0.816	0.634	0.836	0.834	0.838	0.836	0.673	0.772	0.872	0.757	0.764	0.632
	MultiROCKET	0.822	0.813	0.832	0.824	0.645	0.825	0.831	0.820	0.824	0.651	0.758	0.866	0.747	0.750	0.612
	Hydra	0.846	0.827	0.865	0.849	0.693	0.851	0.840	0.861	0.852	0.702	0.789	0.879	0.769	0.780	0.658
	MultiROCKET-Hydra	0.822	0.809	0.834	0.824	0.644	0.835	0.840	0.830	0.834	0.670	0.759	0.866	0.746	0.750	0.611
Baseline + Base-pair probability (Incorporated) (cfg 4)	ROCKET	0.799	0.771	0.827	0.805	0.600	0.809	0.786	0.832	0.813	0.619	0.737	0.850	0.712	0.724	0.573
	MiniROCKET	0.776	0.756	0.797	0.781	0.554	0.801	0.808	0.794	0.799	0.603	0.705	0.835	0.675	0.684	0.521
	MultiROCKET	0.814	0.801	0.828	0.817	0.630	0.816	0.812	0.820	0.816	0.634	0.726	0.848	0.706	0.712	0.556
	Hydra	0.822	0.787	0.857	0.828	0.647	0.834	0.828	0.840	0.835	0.669	0.759	0.862	0.734	0.746	0.608
	MultiROCKET-Hydra	0.814	0.802	0.820	0.817	0.629	0.820	0.825	0.816	0.819	0.642	0.736	0.853	0.717	0.723	0.874

CD diagrams



cfg 3 is better than cfg 4.

Evaluation Metric

Binary class

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Sp = \frac{TN}{TN + FP}$$

$$Sn = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

Multi class

$$MCC_{multi} = \frac{c \times s - \sum_k^K p_k \times t_k}{\sqrt{(s^2 - \sum_k^K p_k^2) \times (s^2 - \sum_k^K t_k^2)}}$$

Proposed Methods

Think of it as different "datasets"

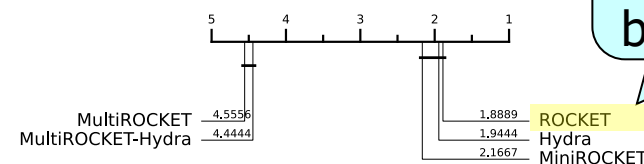
Different transformation methods

Performance across different encoding methods (1/2)

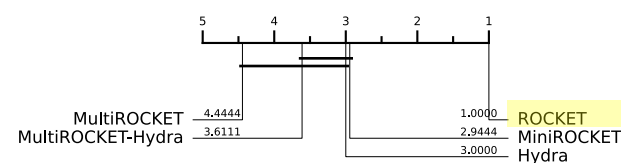
Visualize the "goodness" of a classifier across different "datasets".

	Classifier	5p arm					3p arm					multi-class				
		Acc	Sp	Sn	F1	MCC	Acc	Sp	Sn	F1	MCC	Acc	Sp	Sn	F1	MCC
Single value mapping (enc 1)	ROCKET	0.849	0.842	0.857	0.851	0.699	0.863	0.854	0.873	0.865	0.727	0.853	0.917	0.847	0.851	0.764
	MiniROCKET	0.823	0.809	0.837	0.825	0.647	0.823	0.828	0.817	0.822	0.647	0.835	0.906	0.828	0.833	0.735
	MultiROCKET	0.821	0.802	0.840	0.824	0.643	0.839	0.826	0.852	0.841	0.679	0.811	0.894	0.806	0.809	0.697
	Hydra	0.843	0.820	0.867	0.847	0.688	0.838	0.819	0.857	0.841	0.677	0.831	0.901	0.815	0.827	0.727
	MultiROCKET-Hydra	0.820	0.803	0.837	0.823	0.640	0.840	0.830	0.850	0.841	0.680	0.816	0.896	0.810	0.814	0.704
Grouped variable-length channel mapping (enc 2)	ROCKET	0.835	0.826	0.844	0.836	0.670	0.855	0.849	0.861	0.856	0.710	0.846	0.913	0.839	0.844	0.752
	MiniROCKET	0.843	0.833	0.853	0.844	0.686	0.831	0.821	0.842	0.833	0.663	0.837	0.907	0.828	0.834	0.737
	MultiROCKET	0.819	0.809	0.828	0.820	0.638	0.817	0.814	0.820	0.818	0.634	0.890	0.894	0.806	0.808	0.695
	Hydra	0.825	0.780	0.869	0.832	0.653	0.811	0.769	0.854	0.819	0.626	0.818	0.892	0.765	0.812	0.705
	MultiROCKET-Hydra	0.818	0.814	0.822	0.819	0.636	0.831	0.825	0.837	0.832	0.662	0.820	0.900	0.815	0.818	0.710
Grouped fixed-length channel mapping (enc 3)	ROCKET	0.851	0.843	0.859	0.852	0.702	0.863	0.850	0.875	0.864	0.726	0.849	0.915	0.843	0.847	0.751
	MiniROCKET	0.844	0.836	0.853	0.844	0.686	0.840	0.826	0.855	0.843	0.682	0.851	0.915	0.844	0.849	0.760
	MultiROCKET	0.831	0.815	0.848	0.834	0.663	0.824	0.813	0.836	0.826	0.649	0.811	0.896	0.808	0.808	0.698
	Hydra	0.848	0.816	0.880	0.853	0.699	0.862	0.839	0.884	0.864	0.724	0.843	0.908	0.837	0.839	0.746
	MultiROCKET-Hydra	0.836	0.813	0.859	0.839	0.672	0.833	0.820	0.845	0.835	0.665	0.828	0.905	0.824	0.826	0.728
Cumulative mapping (enc 4)	ROCKET	0.850	0.834	0.866	0.852	0.701	0.863	0.855	0.871	0.864	0.726	0.852	0.915	0.842	0.850	0.762
	MiniROCKET	0.840	0.821	0.860	0.843	0.682	0.840	0.837	0.844	0.841	0.682	0.843	0.911	0.833	0.840	0.747
	MultiROCKET	0.822	0.809	0.834	0.824	0.644	0.832	0.830	0.834	0.832	0.665	0.820	0.898	0.810	0.816	0.709
	Hydra	0.848	0.819	0.878	0.853	0.698	0.853	0.856	0.869	0.855	0.705	0.845	0.910	0.830	0.841	0.749
	MultiROCKET-Hydra	0.824	0.811	0.856	0.825	0.647	0.838	0.833	0.843	0.839	0.677	0.821	0.898	0.810	0.817	0.711
Cumulative grouped variable-length channel mapping (enc 5)	ROCKET	0.843	0.821	0.866	0.847	0.688	0.856	0.840	0.871	0.857	0.712	0.855	0.916	0.843	0.851	0.766
	MiniROCKET	0.845	0.826	0.865	0.848	0.691	0.836	0.833	0.838	0.836	0.672	0.840	0.909	0.833	0.838	0.742
	MultiROCKET	0.826	0.814	0.838	0.828	0.653	0.815	0.820	0.810	0.814	0.631	0.826	0.902	0.820	0.824	0.721
	Hydra	0.850	0.819	0.880	0.854	0.701	0.834	0.807	0.861	0.838	0.669	0.833	0.903	0.818	0.829	0.731
	MultiROCKET-Hydra	0.824	0.810	0.838	0.826	0.649	0.833	0.833	0.833	0.833	0.666	0.830	0.903	0.821	0.827	0.726
Cumulative grouped fixed-length channel mapping (enc 6)	ROCKET	0.856	0.836	0.876	0.858	0.712	0.870	0.861	0.879	0.871	0.741	0.863	0.921	0.852	0.860	0.780
	MiniROCKET	0.856	0.837	0.874	0.858	0.712	0.842	0.839	0.845	0.843	0.685	0.845	0.912	0.837	0.843	0.751
	MultiROCKET	0.820	0.802	0.839	0.824	0.642	0.798	0.798	0.798	0.798	0.597	0.809	0.894	0.806	0.807	0.694
	Hydra	0.850	0.814	0.885	0.855	0.701	0.855	0.840	0.860	0.857	0.711	0.847	0.910	0.831	0.843	0.752
	MultiROCKET-Hydra	0.820	0.801	0.839	0.823	0.641	0.807	0.813	0.802	0.806	0.615	0.821	0.900	0.817	0.819	0.713
Global Cumulative mapping (enc 7)	ROCKET	0.850	0.834	0.866	0.852	0.701	0.863	0.855	0.871	0.864	0.726	0.852	0.915	0.842	0.850	0.762
	MiniROCKET	0.847	0.832	0.862	0.849	0.695	0.848	0.839	0.857	0.850	0.697	0.845	0.911	0.836	0.843	0.750
	MultiROCKET	0.827	0.819	0.834	0.828	0.653	0.847	0.842	0.853	0.848	0.695	0.825	0.901	0.817	0.822	0.718
	Hydra	0.851	0.821	0.880	0.855	0.703	0.861	0.848	0.874	0.863	0.722	0.847	0.911	0.834	0.844	0.753
	MultiROCKET-Hydra	0.829	0.823	0.834	0.830	0.658	0.843	0.838	0.849	0.844	0.688	0.832	0.905	0.823	0.829	0.730
Global Cumulative grouped variable-length channel mapping (enc 8)	ROCKET	0.840	0.814	0.867	0.844	0.682	0.853	0.838	0.867	0.854	0.706	0.856	0.917	0.845	0.853	0.768
	MiniROCKET	0.848	0.834	0.862	0.850	0.697	0.841	0.824	0.839	0.844	0.683	0.844	0.911	0.856	0.842	0.748
	MultiROCKET	0.834	0.828	0.839	0.834	0.668	0.831	0.821	0.842	0.833	0.663	0.828	0.904	0.823	0.826	0.724
	Hydra	0.857	0.821	0.894	0.862	0.717	0.822	0.786	0.857	0.828	0.645	0.826	0.898	0.806	0.820	0.717
	MultiROCKET-Hydra	0.827	0.834	0.839	0.837	0.674	0.834	0.827	0.840	0.835	0.668	0.835	0.907	0.828	0.832	0.734
Global Cumulative grouped fixed-length channel mapping (enc 9)	ROCKET	0.856	0.836	0.876	0.858	0.712	0.870	0.861	0.879	0.871	0.741	0.863	0.921	0.852	0.860	0.780
	MiniROCKET	0.857	0.845	0.870	0.859	0.715	0.840	0.821	0.859	0.843	0.681	0.844	0.911	0.837	0.842	0.749
	MultiROCKET	0.829	0.825	0.833	0.830	0.658	0.820	0.816	0.823	0.820	0.640	0.819	0.900	0.816	0.817	0.710
	Hydra	0.856	0.817	0.894	0.861	0.713	0.859	0.838	0.880	0.862	0.719	0.846	0.911	0.832	0.843	0.752
	MultiROCKET-Hydra	0.829	0.824	0.834	0.830	0.658	0.822	0.825	0.819	0.821	0.644	0.827	0.904	0.823	0.824	0.722

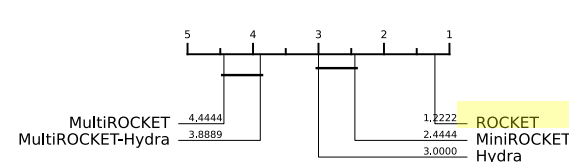
CD diagrams



(a) 5p arm



(b) 3p arm



(c) multi-class

ROCKET is generally the best

MiniROCKET is the fastest

For the downstream analysis, we would use these two for accuracy and efficiency.

Performance across different encoding methods (2/2)

5p arm

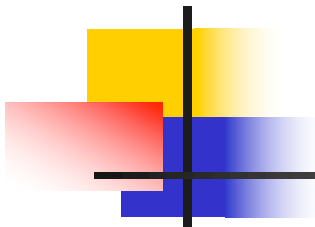
Global	ROCKET	0.856	0.836	0.876	0.858	0.712
Cumulative grouped	MiniROCKET	0.857	0.845	0.870	0.859	0.715
fixed-length	MultiROCKET	0.829	0.825	0.833	0.830	0.658
channel mapping	Hydra	0.856	0.817	0.894	0.861	0.713
(enc 9)	MultiROCKET-Hydra	0.829	0.824	0.834	0.830	0.658

3p arm

Global	0.870	0.861	0.879	0.871	0.741	Global	0.863	0.855	0.871	0.864	0.726
Cumulative grouped	0.840	0.821	0.859	0.843	0.681	Cumulative mapping	0.848	0.839	0.857	0.850	0.697
fixed-length	0.820	0.816	0.823	0.820	0.640	(enc 7)	0.847	0.842	0.853	0.848	0.695
channel mapping	0.859	0.838	0.880	0.862	0.719		0.861	0.848	0.874	0.863	0.722
(enc 9)	0.822	0.825	0.819	0.821	0.644		0.843	0.838	0.849	0.844	0.688

multi-class

Global	0.863	0.921	0.852	0.860	0.780	Grouped	0.849	0.915	0.843	0.847	0.757
Cumulative grouped	0.844	0.911	0.837	0.842	0.749	fixed-length	0.851	0.915	0.844	0.849	0.760
fixed-length	0.819	0.900	0.816	0.817	0.710	channel mapping	0.811	0.896	0.808	0.808	0.698
channel mapping	0.846	0.911	0.832	0.843	0.752	(enc 3)	0.843	0.908	0.837	0.839	0.746
(enc 9)	0.827	0.904	0.823	0.824	0.722		0.828	0.905	0.824	0.826	0.725



Comparison with SOTA

Dataset	Methods	Acc	Sp	Sn	F1	MCC	Time (s)
5p arm	enc 9 + MiniROCKET	0.857	0.845	0.870	0.859	0.715	0.787
	DiCleave	0.818	0.790	0.846	0.822	0.653	21.249
3p arm	enc 9 + ROCKET	0.870	0.861	0.879	0.871	0.741	4.311
	enc 7 + MiniROCKET	0.848	0.839	0.857	0.850	0.697	0.989
	DiCleave	0.854	0.891	0.817	0.847	0.715	15.919
multi-class	enc 9 + ROCKET	0.863	0.921	0.852	0.860	0.780	12.208
	enc 3 + MiniROCKET	0.851	0.915	0.844	0.849	0.760	4.550
	DiCleave	0.820	0.895	0.804	0.815	0.710	131.151

Here, we report the best combination

Here, we report the best combination with respect to MiniROCKET

In general, enc 7/9 is the best encoding method and ROCKET is the best classifier